LIGHTING

Lighting the Stage: Lanterns, Angles and Colour

A ZikShare teaching unit, written for this shelf

What each lantern does, where to hang it, and what light from the front, the side, the back and the top actually does to an actor's face.

WHAT THIS UNIT COVERS

- lantern types
- angles
- colour
- intensity
- safety

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The instruments

Four instruments cover most small venues. The fresnel gives a soft-edged pool and is used for general washes and area light. The profile (an ellipsoidal reflector spotlight) gives a hard edge and takes shutters and gobos, so it is used for specials, for isolating an area and for pattern. The par can gives a very wide, very bright wash with no control of the beam shape, useful for colour and for music. The LED wash gives colour without gels, runs cold, and is now cheaper to run than tungsten, though its dimming curve is different and cheap units buzz.

Know three numbers for each instrument: the beam angle, the throw distance to the stage, and the wattage. A 500 W fresnel at six metres lights a pool about two metres across and will not light a whole stage. Design from the throw you have, not the throw you want.

Angles, and what they say

Front light (about 45 degrees above and 45 degrees to the side) is the standard "open" light: it shows the face, fills the eye sockets and makes the actor look well. Flat front light at eye level flattens the face and kills the set.

Side light rakes across the face and separates the actor from the background, which is why dance and musical numbers are lit from the wings. Back light outlines the body and creates depth; without it, actors stand in front of a flat wall of colour. Top light throws the eye sockets into shadow and reads as sinister or as outdoors at noon. Underlight, from below the face, is the oldest horror effect there is and should be used once.

A useful rule for a scene: one angle for the area, a second angle at 90 degrees for shape, and a back light for separation. Three instruments per area will light most small stages convincingly.

Colour

Gel colour is subtractive: every gel cuts light, and a deep blue may cut 70 per cent of it. Check the transmission figure before you plan a dark blue wash with two lanterns.

Warm front light (straw, pale amber) and cool back light (steel blue) is the standard combination because it reads as sunlight and shadow at once. Colour on the face must stay close to skin tone or the actor stops looking human; colour belongs in the back light, the set and the floor.

Safety and the boring parts

Lanterns get hot, cables trip people, and a badly rigged bar falls. Use safety bonds on every lantern, keep cables taped or matted, do not exceed the circuit loading, and never work at height alone. The lighting designer's first duty is that nobody is hurt by the lighting.

Power, circuits and the venue check

Before you design anything, walk into the venue and ask the electrician four questions: how many circuits are there, what is the rating of each, where is the dimmer rack, and what is the earthing arrangement? A lantern is not a lamp you plug into a wall socket; the load has to be calculated and spread. Six 500 W lanterns on one 10 A circuit will trip it, and a tripped circuit in the middle of a performance is a blackout with a cause.

Count the outlets, the bars and the positions. Many Nigerian halls have one lighting bar, no dimmer, and a single switch by the door; a design for that venue is a design for switching, not for dimming, and it is built from parcans,

floor stands and practicals. Design for the room you are in.

Check the safety items before the first hang: safety bonds, cable ramps, fire extinguishers, and a ladder that is not a chair. Write a one-page risk note for the get-in and give it to the stage manager. Nobody remembers the design that was beautiful; everybody remembers the one that hurt someone.

The lighting states a play actually needs

Most plays need six or seven states, not sixty: a general warm wash, a cooler general, a night state, a special for one area, a practical state for a lamp or a fire, a blackout, and one state that is specific to the play and cannot be described in advance. Build those first and the design will hold the evening.

Resist the temptation to light every moment differently. Constant change exhausts the audience and the operator, and it flattens the two or three moments that should be different. A lighting design is a hierarchy of attention, and hierarchy means most of the stage is in the same light most of the time.

Exercises

1. Light one actor with three lanterns only: front, side, back. Photograph each combination and label what each angle contributes.
2. Take a gel swatch book and find three gels that read as "moonlight" and three as "lamplight". Note their transmission figures.
3. Measure the throw from your rigging position to the stage centre and calculate which lens angle covers your acting area.

Where to read more

Francis Reid, The Stage Lighting Handbook (6th ed.)

Richard Pilbrow, Stage Lighting Design (1997)

J. Michael Gillette, Designing with Light (2008)

These are books to look for in a library or a bookshop. They are not on this shelf and ZikShare does not host them.